

24.2 PET Scanning

Question Paper

Course	CIEA Level Physics
Section	24. Medical Physics
Topic	24.2 PET Scanning
Difficulty	Medium

Time allowed: 50

Score: /39

Percentage: /100

Question 1a

Positron emission tomography (PET scanning) makes use of a tracer containing a radioactive material that decays by positron emission.

(i)

State what is meant by a tracer.

[2]

(ii)

State the name of the particles that are emitted from the body and detected by the detectors during PET scanning.

[1]

[3 marks]

Question 1b

Explain how the particles in **(a)(ii)** are created from positrons.

[3 marks]

Question 1c

Positrons can be artificially created by a process in the laboratory that is the reverse of the process in **(b)**. This process creates both a positron and an electron moving at the same speed in opposite directions.

Suggest why two of the particles in **(a)(ii)** are needed to create one positron.

[2 marks]

Question 2a

Positron emission tomography (PET scanning) obtains diagnostic information from a person. The information is used to form an image.

PET scanning uses a tracer and involves annihilation.

Explain what is meant by

(i)
a tracer,

[1]

(ii)
annihilation.

[1]

[2 marks]

Question 2b

For the annihilation process

(i)
State the names of the particles involved.

[1]

(ii)
Calculate the total energy released in one annihilation event.

[2]

(iii)
Calculate the wavelength of each gamma photon released.

[2]

[5 marks]

Question 2c

State and explain the direction, relative to each other, in which the gamma photons are emitted.

[2 marks]

Question 2d

Explain how the gamma photons are used to produce an image.

[3 marks]

Question 3a

Positron emission tomography (PET) is useful in medical diagnosis. PET can be used to locate tumours or areas of increased activity within the brain.

(i)

Describe the principles of PET.

[4]

(ii)

Explain how PET scans are able to locate tumours or areas of increased activity within the brain.

[2]

[6 marks]**Question 3b**

The decay of the nuclei in the tracer material results in the emission of a positron. Two of the detectors, directly opposite to each other, are triggered as they each receive a gamma photon from the annihilation of the positron.

Suggest why the annihilation

(i)
occurs close to the point where the positron is emitted,

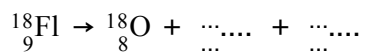
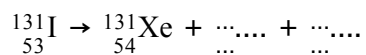
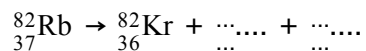
[2]

(ii)
results in two gamma-ray photons of equal energy moving in opposite directions.

[2]**[4 marks]**

Question 3c

Three tracer isotopes, rubidium-82, fluorine-18 and iodine-131, are being considered for use in a PET scanner. The incomplete nuclear decay equations are shown below.



(i)

Complete the nuclear decay equations for rubidium-82, iodine-131 and fluorine-18.

[3]

(ii)

State and explain which tracer would **not** be suitable for use in a PET scanner.

[2]

[5 marks]

Question 3d

It is suggested that a scanner could be designed to detect photons produced from the annihilation of a proton and an antiproton.

Calculate:

(i)
the energy released in the proton–antiproton annihilation. [2]

(ii)
the wavelength of the γ -ray photons produced in the annihilation. [2]

[4 marks]



Model Answers: Medium

1a

(a)

(i) A tracer is:

- A substance introduced into the body; [1 mark]
- That is absorbed by tissues that are being studied; [1 mark]

(ii) The particles that are emitted from the body and detected by the detectors during PET scanning are:

- Gamma photons; [1 mark]

[Total: 3 marks]

1b

(b) Gamma photons are created from positrons as:

- The positrons emitted from the tracer in the body interact with electrons in the body tissue; [1 mark]
- The positron is an antiparticle of the electron; [1 mark]
- So, pair annihilation takes place between the positron and the electron / the matter turns into energy; [1 mark]

[Total: 3 marks]

1c

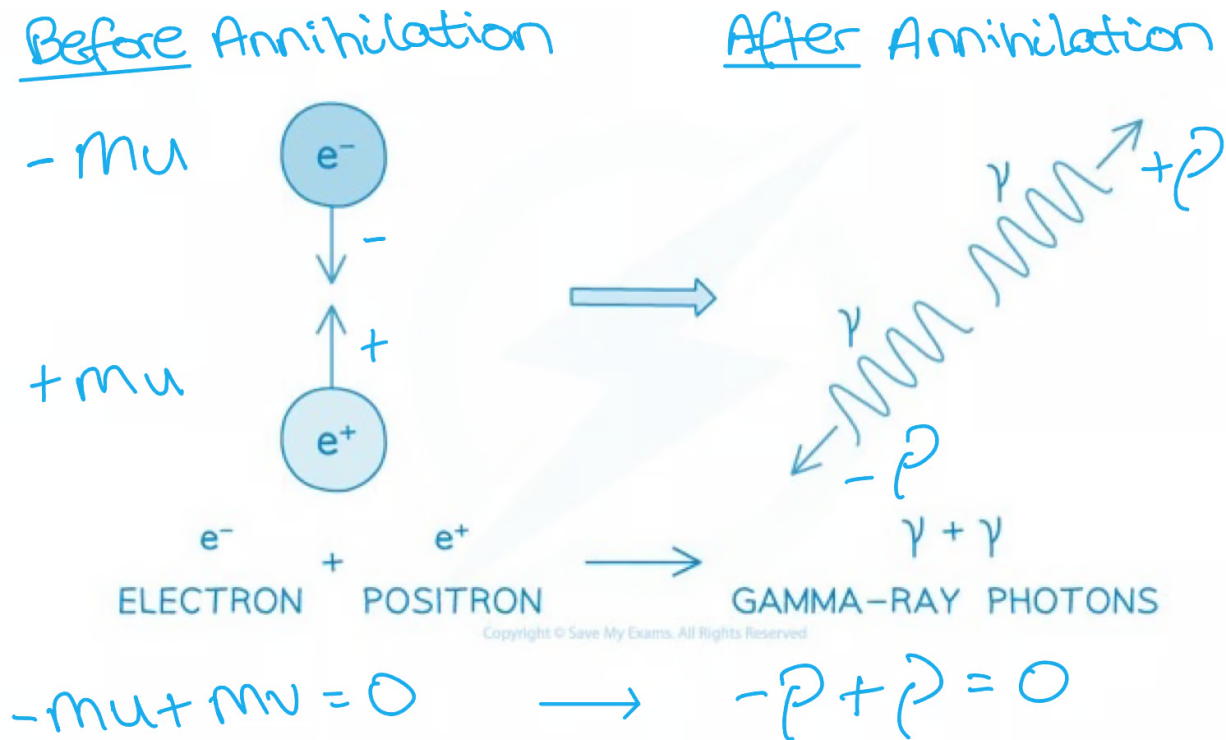
(c) Two gamma photons are needed to create one positron because:

- (The positron and the electron have the) same mass and travel in opposite directions, so their combined momentum before annihilation is zero; [1 mark]

- Hence, two gamma photons must be produced travelling in opposite directions (to conserve momentum after annihilation); [1 mark]

[Total: 2 marks]

- The concept of **conservation of momentum** before and after annihilation is tricky:



2a

(a)

(i) A tracer is:

- A substance containing radioactive nuclei that are introduced into the body / absorbed by the tissue being studied; [1 mark]

(ii) Annihilation is:

- When a particle collides with its antiparticle so that mass is converted into energy; [1 mark]

[Total: 2 marks]

2b

(b)

(i) The names of the particles involved in annihilation are:

- Electron(s) and positron(s); [1 mark]

(ii) Calculate the total energy released in one annihilation event:

List the known quantities:

- Mass of an electron/positron, $m_e = 9.11 \times 10^{-31} \text{ kg}$
- Speed of light, $c = 3.00 \times 10^8 \text{ m s}^{-1}$
- Planck's constant, $h = 6.63 \times 10^{-34} \text{ J s}$

Use mass-energy equivalence to determine the total energy:

- Mass-energy equivalence: $E = mc^2$
- Total energy: $E = 2m_e c^2$ [1 mark]
- Total energy: $E = 2 \times (9.11 \times 10^{-31}) \times (3.00 \times 10^8)^2 = 1.64 \times 10^{-13} \text{ J}$ [1 mark]

(iii) Calculate the wavelength of each gamma photon released:

- Photon energy: $E = \frac{hc}{\lambda}$
- Photon wavelength: $\lambda = \frac{2hc}{E} = \frac{2 \times (6.63 \times 10^{-34}) \times (3.00 \times 10^8)}{1.64 \times 10^{-13}}$ [1 mark]
- Photon wavelength: $\lambda = 2.43 \times 10^{-12} \text{ m}$ [1 mark]

[Total: 5 marks]

2c

(c) The gamma photons are emitted:

- In opposite directions; [1 mark]

Because:

Max **one** from:

- The total momentum (before and after interaction) is zero; [1 mark]
- Total momentum must be conserved (in the interaction); [1 mark]
- The momentum of the photons must be equal and opposite; [1 mark]

[Total: 2 marks]

2d

(d) The gamma photons produce an image because:

Any **three** from:

- Gamma photons are detected (outside the body / by detectors); [1 mark]
- Gamma photons arrive (at the detector) at different times; [1 mark]
- The location of production (of the gamma photons) is detected / determined; [1 mark]
- An image of the tracer concentration in tissue is produced; [1 mark]

[Total: 3 marks]

3a

(a)

(i) The principles of PET are:

Any **four** from:

- A positron / beta-plus emitting tracer / source is used; [1 mark]
- The positrons annihilate with electrons (inside the patient); [1 mark]
- Positron-electron annihilation produces two gamma photons; [1 mark]
- The gamma photons travel in opposite directions; [1 mark]
- The patient is surrounded by a ring of gamma detectors; [1 mark]
- A 3-D image is created (by the computer connected to the detectors); [1 mark]

- The total momentum (before and after interaction) is zero; [1 mark]
- Total momentum must be conserved (in the interaction); [1 mark]
- The momentum of the photons must be equal and opposite; [1 mark]

[Total: 2 marks]

2d

(d) The gamma photons produce an image because:

Any **three** from:

- Gamma photons are detected (outside the body / by detectors); [1 mark]
- Gamma photons arrive (at the detector) at different times; [1 mark]
- The location of production (of the gamma photons) is detected / determined; [1 mark]
- An image of the tracer concentration in tissue is produced; [1 mark]

[Total: 3 marks]

3a

(a)

(i) The principles of PET are:

Any **four** from:

- A positron / beta-plus emitting tracer / source is used; [1 mark]
- The positrons annihilate with electrons (inside the patient); [1 mark]
- Positron-electron annihilation produces two gamma photons; [1 mark]
- The gamma photons travel in opposite directions; [1 mark]
- The patient is surrounded by a ring of gamma detectors; [1 mark]
- A 3-D image is created (by the computer connected to the detectors); [1 mark]

(ii) PET scans are able to locate tumours or areas of increased activity within the brain because:

- The pair of photons are produced at the same time and travel to (diametrically) opposite detectors; [1 mark]
- The arrival times of the photons / delay time indicates the location (of tumour or

areas of increased activity); [1 mark]

[Total: 6 marks]

3b

(b)

(i) The annihilation occurs close to the point where the positron is emitted because:

- Positrons annihilate with electrons, which are in all atoms (in the patient); [1 mark]
- Therefore, the positron will not travel far before encountering an electron; [1 mark]

(ii) The annihilation results in two gamma photons of equal energy moving in opposite directions because:

- Before the annihilation, the total momentum is zero
- (Therefore) for momentum to be conserved, the total momentum of the photons must equal zero; [1 mark]
- (Hence) if they have equal momenta, they must have equal photon energies; [1 mark]

[Total: 4 marks]

Remember that the momentum of a photon is $p = \frac{E}{c}$, therefore, the momentum p is directly proportional to energy E .

3c

(c)

(i) The complete nuclear decay equations are:

- ${}_{37}^{82}\text{Rb} \rightarrow {}_{36}^{82}\text{Kr} + {}_{+1}^0\beta + {}_0^0\gamma$ [1 mark]
- ${}_{53}^{131}\text{I} \rightarrow {}_{54}^{131}\text{Xe} + {}_{-1}^0\beta + {}_0^0\bar{\nu}$ [1 mark]
- ${}_{9}^{18}\text{F} \rightarrow {}_{8}^{18}\text{O} + {}_{+1}^0\beta + {}_0^0\gamma$ [1 mark]

(ii) The tracer that would **not** be suitable for use in a PET scanner is:

- Iodine-131 **AND** because it is a beta-minus emitter; [1 mark]
- Therefore, it would not annihilate with electrons in the patient's body; [1 mark]

[Total: 5 marks]

Remember that beta-minus particles are **electrons** whilst beta-plus particles are **positrons**. The tracer needs to emit positrons for them to annihilate with the electrons in the body.

3d

(d)

List the known quantities:

- Mass of a proton/antiproton, $m_p = 1.67 \times 10^{-27} \text{ kg}$
- Speed of light, $c = 3.00 \times 10^8 \text{ m s}^{-1}$
- Planck's constant, $h = 6.63 \times 10^{-34} \text{ J s}$

Use mass-energy equivalence to determine the total energy:

- Mass-energy equivalence: $E = mc^2$
- Total energy: $E = 2m_p c^2$ [1 mark]
- Total energy: $E = 2 \times (1.67 \times 10^{-27}) \times (3.00 \times 10^8)^2 = 3.0 \times 10^{-10} \text{ J}$ [1 mark]

(iii) Calculate the wavelength of each gamma photon released:

- Photon energy: $E = \frac{hc}{\lambda}$
- Photon wavelength: $\lambda = \frac{2hc}{E} = \frac{2 \times (6.63 \times 10^{-34}) \times (3.00 \times 10^8)}{3.0 \times 10^{-10}}$ [1 mark]
- Photon wavelength: $\lambda = 1.32 \times 10^{-15} \text{ m}$ [1 mark]

[Total: 4 marks]